

Node.js & Express Bootcamp – Module 1 Exercise

Hands-on Exercise No. 1

Course: Node.js & Express Bootcamp

Module: Node.js Fundamentals

Total Marks: 10

Due Date: 22/05/2026

Instructions:

- Prepare your solution in MS Word
 - Submit in .doc or .docx format
 - Late or copied submissions will not be accepted
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Learning Outcome:

- Understand Node.js runtime and architecture
 - Work with modules and NPM packages
 - Handle files using File System module
 - Understand Event Loop behavior
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Problem Statement:

You are working as a Backend Developer for a Content Management System. The client needs a simple Node.js prototype to demonstrate file handling and event-driven programming concepts.

Task Requirements:

- Create a new Node.js project
- Initialize project using npm init
- Install at least one package (e.g. chalk for colored console output)
- Use File System module to:
 - Create a file named students.txt
 - Write sample student names into the file
 - Read and display file contents in console

- Demonstrate Event Loop behavior by combining:
 - Synchronous code
 - Asynchronous code (e.g. setTimeout)
 - Take screenshots of:
 - Project setup and npm installation
 - File creation and reading output
 - Console output showing Event Loop execution
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Submission Guidelines:

- Upload your .docx file on LMS
- Include:
 - Code snippets
 - Console outputs
 - Screenshots of execution steps
- Add short explanations for each feature